DEPARTMENT OF ELECTRICAL & ELECTRONIC ENGINEERING

Experiment No. : 3

Experiment Name: Stability Analysis in Routh- Hurwitz criteria using MATLAB

OBJECTIVES

1.1 Understanding the basic principle of Routh-Hurwitz criteria

1.2 Using MATLAB program analyze the stability & compare with theoretical calculation.

PRELAB WORK:

1. **Read this laboratory tutorial carefully before coming to the laboratory class, so that you know what is required.**
2. Try to follow the lectures of EEE 4701.
3. Familiarize yourself with the relevant MATLAB functions and codes necessary for this experiment.

OBJECTIVE 1.1 Understanding the basic principle of Routh-Hurwitz criteria

Introduced in 1874

- Generalizes the previous method
- Introduces additional combinations of coefficients
- Based on Sturm's theorem.

Central is the idea of the "Routh Table"

Step 1: Write the polynomial as

$$d(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$$

Step 2

Write the coefficients in 2 rows

- First row starts with a_n
- Second row starts with a_{n-1}
- Other coefficients alternate between rows
- Both rows should be same length
  - Continue until no coefficients are left
  - Add zero as last coefficient if necessary

TABLE 6.1 Initial layout for Routh table

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2			
s^1			
s^0			

Step 3

Complete the third row.

- Call the new entries $b_1, \cdots, b_k$
  - The third row will be the same length as the first two

$$b_1 = -\frac{\det\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} \qquad b_2 = -\frac{\det\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} \qquad b_3 = -\frac{\det\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3}$$

- The denominator is the first entry from the previous row.
- The numerator is the determinant of the entries from the previous two rows:
  - The first column
  - The next column following the coefficient

$$b_k = -\frac{\det\begin{vmatrix} a_n & a_{n-2k} \\ a_{n-1} & a_{n-2k-1} \end{vmatrix}}{a_{n-1}}$$

  - If a coefficient doesn't exist, substitute 0.

Step 4

TABLE 6.2 Completed Routh table

s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2	$\frac{-\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$	$\frac{-\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$	$\frac{-\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$
s^1	$\frac{-\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$	$\frac{-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$	$\frac{-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$
s^0	$\frac{-\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$	$\frac{-\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$	$\frac{-\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$

Treat each following row in the same way as the third row

- There should be $n+1$ rows total, including the first row.

Step 4

Now examine the first column

TABLE 6.3 Completed Routh table for Example 6.1

s^3	1	31	0
s^2	$\cancel{10}\ 1$	$\cancel{1030}\ 103$	0
s^1	$\dfrac{-\begin{vmatrix} 1 & 31 \\ 1 & 103 \end{vmatrix}}{1} = -72$	$\dfrac{-\begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix}}{1} = 0$	$\dfrac{-\begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}}{1} = 0$
s^0	$\dfrac{-\begin{vmatrix} 1 & 103 \\ -72 & 0 \end{vmatrix}}{-72} = 103$	$\dfrac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$	$\dfrac{-\begin{vmatrix} 1 & 0 \\ -72 & 0 \end{vmatrix}}{-72} = 0$

Theorem 1.

The number of sign changes in the first column of the Routh table equals the number of roots of the polynomial in the Closed Right Half-Plane (CRHP).

Note: Any row can be multiplied by any positive constant without changing the result.

1.2 Objective: Using MATLAB program analyze the stability & compare with theoretical calculation.

Below here there is a MATLAB program using RH criteria

```
%% routh hurwitz criteria
clear
clc
%% firstly it is required to get first two row of routh matrix
e=input('enter the coefficients of characteristic equation: ');
disp('-------------------------------------------------------------------------')
l=length(e);
m=mod(l,2);
if m==0
   a=rand(1,(l/2));
   b=rand(1,(l/2));
   for i=1:(l/2)
      a(i)=e((2*i)-1);
      b(i)=e(2*i);
   end
else
   e1=[e 0];
   a=rand(1,((l+1)/2));
   b=[rand(1,((l-1)/2)),0];
   for i=1:((l+1)/2)
      a(i)=e1((2*i)-1);
      b(i)=e1(2*i);
```

```
    end
end
%% now we genrate the remaining rows of routh matrix
l1=length(a);
c=zeros(l,l1);
c(1,:)=a;
c(2,:)=b;
for m=3:l
    for n=1:l1-1
        c(m,n)=-(1/c(m-1,1))*det([c((m-2),1) c((m-2),(n+1));c((m-1),1) c((m-1),(n+1))]);
    end
end
disp('the routh matrix:')
disp(c)
%% now we check the stablity of system
if c(:,1)>0
    disp('System is Stable')
else
    disp('System is Unstable');
end
```

Task: Use Characteristic Equation :

$ID*s^4 + (ID-1)* s^3 + (ID-2)* s^2 +(ID-3)* s + ID-4=0$ [ID=even]

$ID*s^5+(ID-1)*s^4 + (ID-2)* s^3 + (ID-3)* s^2 +(ID-4)* s + ID-5=0$ [ID= odd]

Using **Routh-Hurwitz criteria,** find out if the system is stable or not. Use the last two digits of your ID.

Report preparation

Please prepare your report with yourself & don't try to copy. Report should cover the following Topics:

1. Objective
2. Apparatus/Utility software:
3. Description of Task:
4. MATLAB code & Result:
5. Analysis of Each output:
6. Discussion: